Exercise 1 — Mean Calculation

Employee Salaries:

25000, 30000, 35000, 40000, 45000

1. Mean

Formula:

$$Mean = \frac{\sum x}{n}$$

$$= \frac{25000 + 30000 + 35000 + 40000 + 45000}{5}$$

$$= \frac{175000}{5}$$

$$= 35000$$

Mean = 35,000

2. Median

Arrange data:

25000, 30000, 35000, 40000, 45000

Middle value:

Median = 35,000

3. Mode

Mode = Most frequently occurring value

All numbers appear once.

Mode = No Mode

Exercise 2 — Variance & Standard Deviation

Marks:

60, 70, 80, 90, 100

Step 1: Mean

$$Mean = \frac{60 + 70 + 80 + 90 + 100}{5}$$

$$= \frac{400}{5}$$

$$= 80$$

Step 2: Variance

$$Variance = \frac{\sum (x - \mu)^2}{n}$$

Marks	Difference	Square
60	-20	400
70	-10	100
80	0	0
90	10	100
100	20	400

Total:

$$400 + 100 + 0 + 100 + 400 = 1000$$

$$Variance = \frac{1000}{5}$$

$$= 200$$

Variance = 200

Step 3: Standard Deviation

$$SD = \sqrt{200}$$

$$= 14.14$$

Standard Deviation = 14.14

Explanation

Standard deviation shows **how spread out data values are from the mean value.**

Small SD → data close together

Large SD → data spread apart

Exercise 3 — AI vs ML vs Deep Learning

Scenario	Type
Face recognition system	Deep Learning
Spam email filtering	Machine Learning
Chess playing robot	Artificial Intelligence
Netflix recommendations	Machine Learning

Exercise 4 — Supervised Learning Identification

Problem	Yes / No
Predict house prices	Yes
Customer segmentation	No
Predict exam pass/fail	Yes
Fraud detection with labeled data	Yes

Explanation:

Supervised learning uses **labeled data**.

Exercise 5 — Unsupervised Learning Identification

1. Learning Type

Unsupervised Learning

2. Suggested Algorithm

K-Means Clustering

3. Why labels are not needed?

Labels are not required because the system finds hidden patterns and groups customers automatically based on similarities.

Exercise 6 — Reinforcement Learning Scenario

Robot:

- Rewards for balance
- Penalties for falling

Answers

Learning Type:

Reinforcement Learning

Agent:

Agent is the learner or decision maker (robot).

Reward:

Reward is feedback received after an action.

Positive reward → good action

Negative reward → bad action

Exercise 7 — Classification vs Regression

Problem	Type
Predict salary	Regression
Predict disease yes/no	Classification
Predict stock price	Regression
Email spam detection	Classification

Explanation:

Classification → categorical output

Regression → numerical output

Exercise 8 — K-Means Clustering

Mall data:

- Income
- Spending Score

Answers

Algorithm:

K-Means Clustering

Learning Type:

Unsupervised Learning

Cluster:

Cluster is a group of similar data points having common characteristics.

Exercise 9 — Deep Learning Use Cases

Problem	Model
Face recognition	CNN
Language translation	LSTM
Stock price prediction	RNN
Object detection	CNN

Explanation:

CNN → images

RNN → sequential data

LSTM → long sequence dependencies

Exercise 10 — Real-Time Case Study

Bank tasks:

- Detect fraud transactions
- Analyze customer spending
- Predict loan approvals

1. Supervised Learning

Predict loan approvals

Fraud detection with labeled data

2. Unsupervised Learning

Customer spending analysis

3. Metrics for Fraud Detection

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

4. Algorithm for Loan Approval

- Logistic Regression
- Decision Tree
- Random Forest

Coding Challenge (Google Collab)

Program 1 — Mean and Median

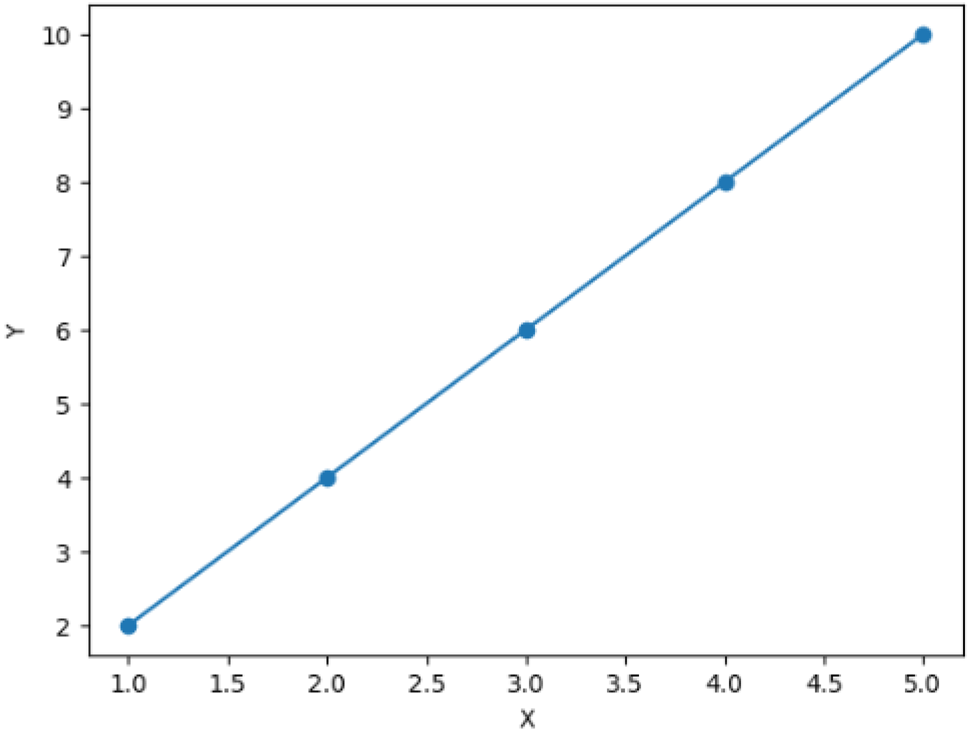
```
import statistics

data = [25000,30000,35000,40000,45000]

mean_value = statistics.mean(data)
median_value = statistics.median(data)

print("Mean:", mean_value)
print("Median:", median_value)
```

```
Mean: 35000
Median: 35000
```

Program 2 — Simple Linear Regression

```python
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

X = np.array([1,2,3,4,5]).reshape(-1,1)
y = np.array([2,4,6,8,10])

model = LinearRegression()
model.fit(X,y)

prediction = model.predict(X)

print("Predictions:")
print(prediction)

plt.scatter(X,y)
plt.plot(X,prediction)

plt.xlabel("X")
plt.ylabel("Y")

plt.show()
```

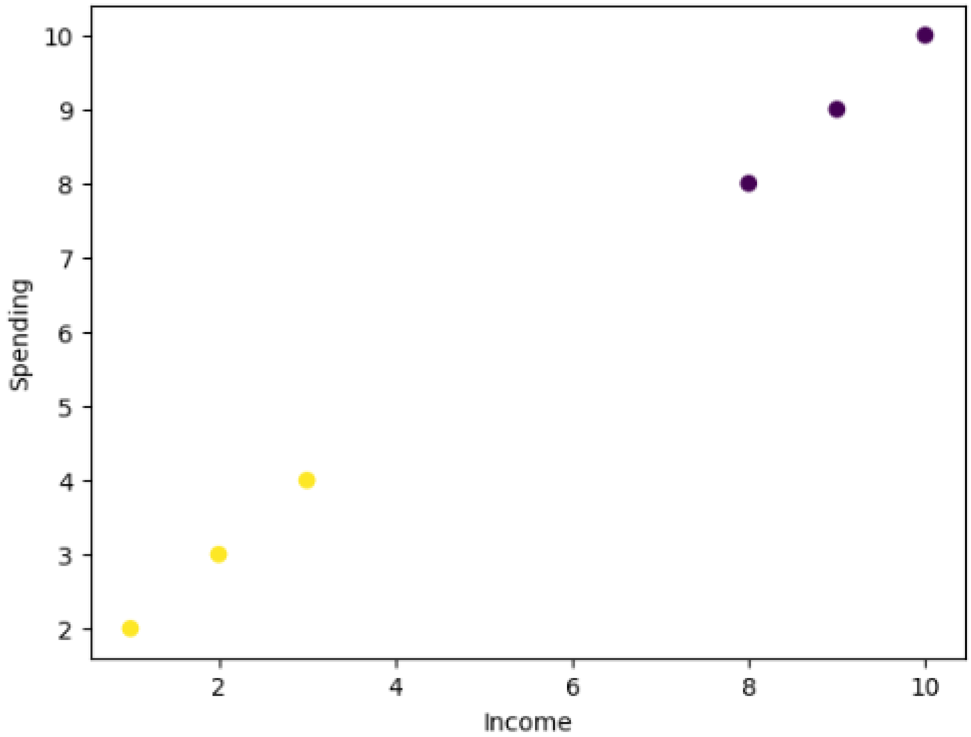
```
Predictions:
[ 2.  4.  6.  8. 10.]
```



Program 3 — K-Means Clustering

```
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

X = np.array([
    [1,2],
    [2,3],
    [3,4],
    [8,8],
    [9,9],
    [10,10]
])

kmeans = KMeans(n_clusters=2)

kmeans.fit(X)

labels = kmeans.labels_

print(labels)

plt.scatter(X[:,0], X[:,1], c=labels)

plt.xlabel("Income")
plt.ylabel("Spending")

plt.show()
```

```
[1 1 1 0 0 0]
```



Program 4 — Logistic Regression

```
from sklearn.linear_model import LogisticRegression
import numpy as np

X = np.array([
    [1],
    [2],
    [3],
    [4]
])

y = np.array([0,0,1,1])

model = LogisticRegression()

model.fit(X,y)

prediction = model.predict([[2.5]])

print("Prediction:", prediction)
```

```
Prediction: [1]
```

Program 5 — Plot Data using Matplotlib

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
y = [10,20,30,40,50]

plt.plot(x,y)

plt.xlabel("X values")
plt.ylabel("Y values")

plt.title("Simple Plot")

plt.show()
```